

Participation Journal

By: Tyler Trotter

Entry 2: (11/14/2021)

Proof Writing Techniques

Since the beginning of the semester, I have spent a lot of time reflecting on proofs for this course. I feel I have grown significantly as a proof-writer compared to the start when I felt rather unconfident on what constituted a proof. Now, I quite enjoy proof writing as it feels like a logic puzzle that is awaiting a clever, interesting, or beautiful solution upon which you can build more and more interesting results. To know something for certain, according to the definitions, is both liberating and exhausting at the same time.

In order to get better at proof writing, I have spent a lot of time meeting with people in the tutoring center who have a better grasp on the material than I do and discussing nearly all of my proofs with them, having them help me achieve a better understanding of where I went wrong when I did. Most importantly, for me, has been doing my proof writing on a white board, letting it sit there and then working through it in my mind, step-by-step, and along the way asking myself the question: "Is this logically correct?" and supposing that it is, I ask: "Does this prove what I am intending to prove?" I often have to re-write my proofs and that is possibly the most important part, because it means somewhere I identified an incorrect statement or failed to show what I was intending to. It is in this step that the most learning takes place—to recognize where you went wrong.

I have found that much of proof writing becomes a matter of understanding thoroughly the theorems which you want to reference and the definitions which you are dealing with. In many cases, if you understand these two things very well the proofs can often write themselves. Therefore to show something rigorously it may be that you show the conditions of a theorem have been satisfied and the result you want to prove follows. Thus, memorization of these theorems has been a key part in this process for me.